Coursework for the course ‘G11PT - Production Technology’ or ‘G11TS – Managing Wells’

This coursework carries 30% of the course mark. The rest of the mark comes from the exam performance.

1. Answer only the questions specified for your student ID number. Answers to any other questions will be disregarded. The next page lists what questions you should answer.

2. Stick to the word limits. Insert as many figures as needed to explain your answer.

3. Your answers should: be developed by you alone; be in your own words; be solely based on the knowledge attained from this course (except the literature review part at the end); not be exchanged or plagiarised; not use Generative AI tools.

4. State clearly any assumptions used.

5. Whenever a sub-question requires you to list or name a specified number of items, any items you list in excess of the number specified may not be marked.

Attached: Appendices with figures

What questions you should answer:

Check your student ID number (most often it looks something like H00123456).

If the last digit in your ID number is:	then you should answer questions:
0	1A, 1D, 1G, 1J, 1K, 1L, 1M, 1N, 2A
1	1A, 1E, 1H, 1J, 1K, 1L, 1M, 1N, 2B
2	1A, 1F, 1I, 1J, 1K, 1L, 1M, 1N, 2C
3	1A, 1F, 1G, 1J, 1K, 1L, 1M, 1N, 2D
4	1B, 1F, 1G, 1J, 1K, 1L, 1M, 1N, 2E
5	1B, 1D, 1H, 1J, 1K, 1L, 1M, 1N, 2A
6	1B, 1E, 1I, 1J, 1K, 1L, 1M, 1N, 2B
7	1C, 1E, 1G, 1J, 1K, 1L, 1M, 1N, 2C
8	1C, 1F, 1H, 1J, 1K, 1L, 1M, 1N, 2D
9	1C, 1D, 1I, 1J, 1K, 1L, 1M, 1N, 2E

Your answers to any other questions will be disregarded.

Part 1 (total of 50 marks)

1.A. Consider the well completion example shown in Figure 1 in Appendix A. All your sub-questions in Part 1 will be referring to this well. Assume that this well should always produce at some fixed tubing head pressure (THP).

(No marks) ***[no answer expected]***

1.B. Consider the well completion example shown in Figure 2 in Appendix A. All your sub-questions in Part 1 will be referring to this well. Assume that this well should always produce at some fixed tubing head pressure (THP).

(No marks) ***[no answer expected]***

1.C. Consider the well completion example shown in Figure 3 in Appendix A. All your sub-questions in Part 1 will be referring to this well. Assume that this well should always produce at some fixed tubing head pressure (THP).

(No marks) ***[no answer expected]***

1.D. This well is producing from a carbonate, undersaturated-oil reservoir with low permeability.

(No marks) ***[no answer expected]***

1.E. This well is producing from a sandstone, saturated-oil reservoir with high permeability.

(No marks) ***[no answer expected]***

1.F. This well is producing from a sandstone, undersaturated-oil reservoir with low permeability and a very thick and heterogeneous production interval.

(No marks) ***[no answer expected]***

1.G. After some time of production, the well performance has changed and it is suspected that emulsion is forming in the well. The wellbore pressure profile has changed accordingly.

(No marks) ***[no answer expected]***

1.H. After some time of production, the well started producing at a higher GOR so its performance and the wellbore pressure profile have changed accordingly.

(No marks) ***[no answer expected]***

1.I. After some time of production, the well started producing higher concentration of sand so its performance and the wellbore pressure profile have changed accordingly.

(No marks) ***[no answer expected]***

1.J. There is a concern that some completion string components of this well can fail due to this change in the well pressure or fluid composition and properties.
i. Suggest ONE such reasonable component (do NOT consider the packer or seal assembly) and explain what failure you mean.
ii. Explain your suggestion and illustrate how the well pressure has changed and may have caused this failure.
iii. Explain what particular part or mechanism of this component could fail.

(4+5+3 marks)

[Total answer word limit for 1.J: 100 words. Diagrams: as many as needed.]

1.K. (Continuing from question 1.J):
i. If the completion string component failed as you suggested, when and how would this problem be detected?
ii. What TWO production logs could locate and confirm the problem with this completion component?
iii. Explain how this log's measurements would be used for this.

(4+6+6 marks)

[Total answer word limit for 1.K: 100 words. Diagrams: as many as needed.]

1.L. Fortunately, the component problem that you suggested did not happen. However, there is another problem: the well's productivity has decreased. It is suspected that this may be due to some kind of formation damage, related to the change in the fluid flow conditions mentioned earlier.
i. Describe ONE reasonable type of formation damage that may have happened.
ii. Explain why it is related to the earlier-mentioned change in the fluid flow conditions.

(3+4 marks)

[Total answer word limit for 1.L: 100 words. Diagrams: as many as needed.]

1.M. Your company is considering dealing with this formation damage by either matrix or fracture stimulation.
i. What stimulation treatment type do you find most appropriate? Explain your answer.
ii. Briefly describe how the treatment should be designed and executed in this well.

(3+6 marks)

[Total answer word limit for 1.M: 100 words. Diagrams: as many as needed.]

1.N. Another option to deal with this formation damage problem is to re-perforate the production interval to improve the inflow performance. Explain what type of the perforation gun, charges, delivery, and conditions you think would be best in your well.

(6 marks)

[Total answer word limit for 1.N: 100 words. Diagrams: as many as needed.]

Part 2 (total of 50 marks)

There are a number of petroleum business and research-related publications in the https://onepetro.org/ database of publications (see instructions below on how to download such papers for free). Please overview in your own words any 3 published field and/or research case studies related to your question topic (see below), making sure that in total (i.e. not necessarily individually) these publications use several (at least 2) models or correlations as specified in your topic. Compare these model/correlation performances, problems and successes between these cases. Try to explain the differences in the performances. Propose your 'optimal conditions' for one of the models/correlations you have described, based on this small review study, and justify your choice. Feel free to copy the key figures from these publications to your report as required.

Provide a numbered list of the references (i.e. the publications that you have overviewed) at the end of the report. In the report, refer to the references by their numbers in square brackets (e.g. '[2] reported .... The results in [1] showed that...'). The list of references does not count into the word limit.

(50 marks)

[Answer word limit for Part 2: 1000 words. Diagrams: as many as needed.]

2.A. Multiphase flow correlations in horizontal and inclined pipes.
2.B. Gas condensate IPR models.
2.C. Horizontal well IPR models.
2.D. Multilateral well IPR models.
2.E. Multiphase choke flow models.

How to download/read the onepetro.org papers for free:

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Make sure you are a responsible user (when unusually large numbers of SPE papers are downloaded from a student account the SPE or University may consider a penalty).

END OF COURSEWORK TASKS

Appendix A (diagrams related to Part 1)

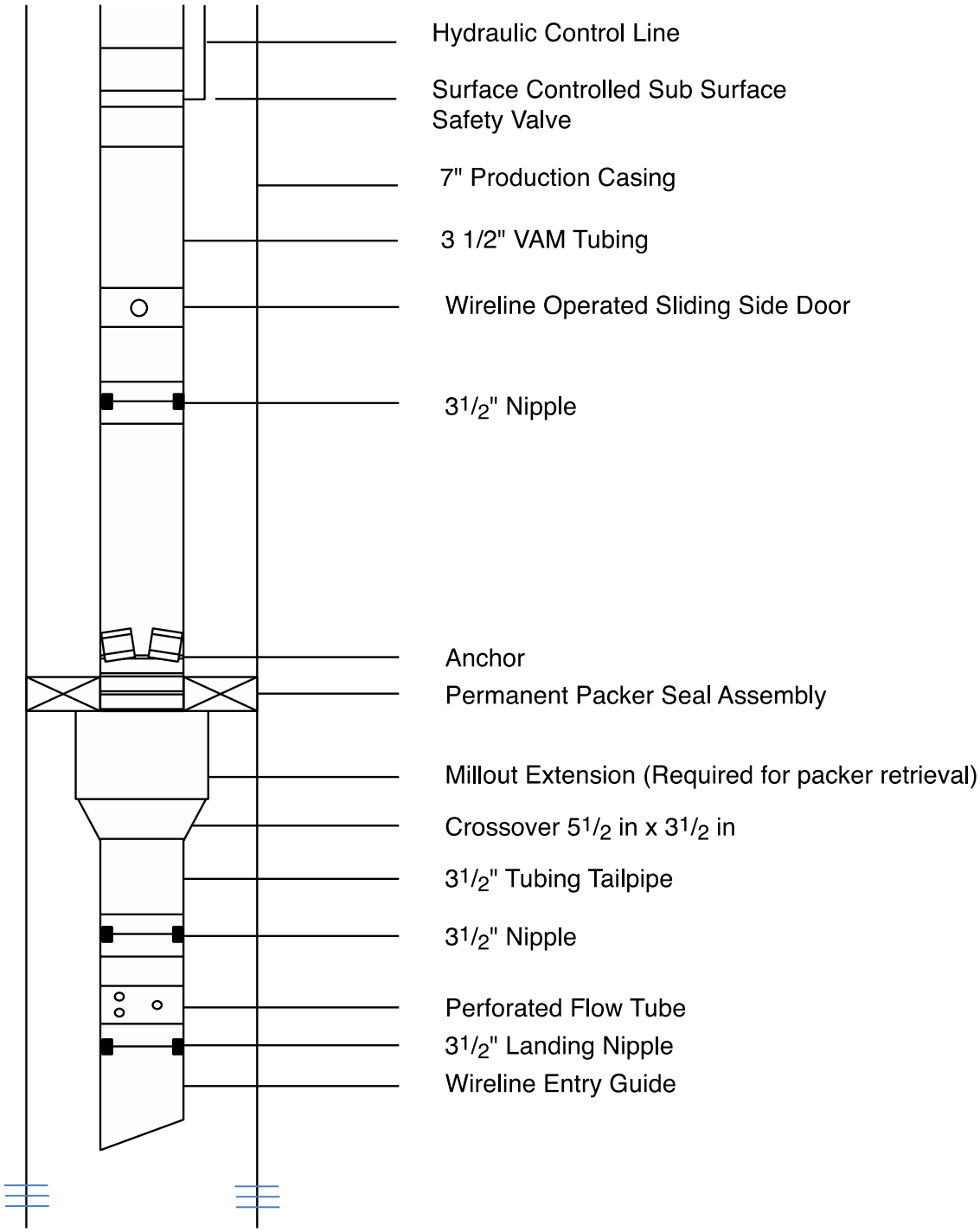

Figure 1: Well completion example 1

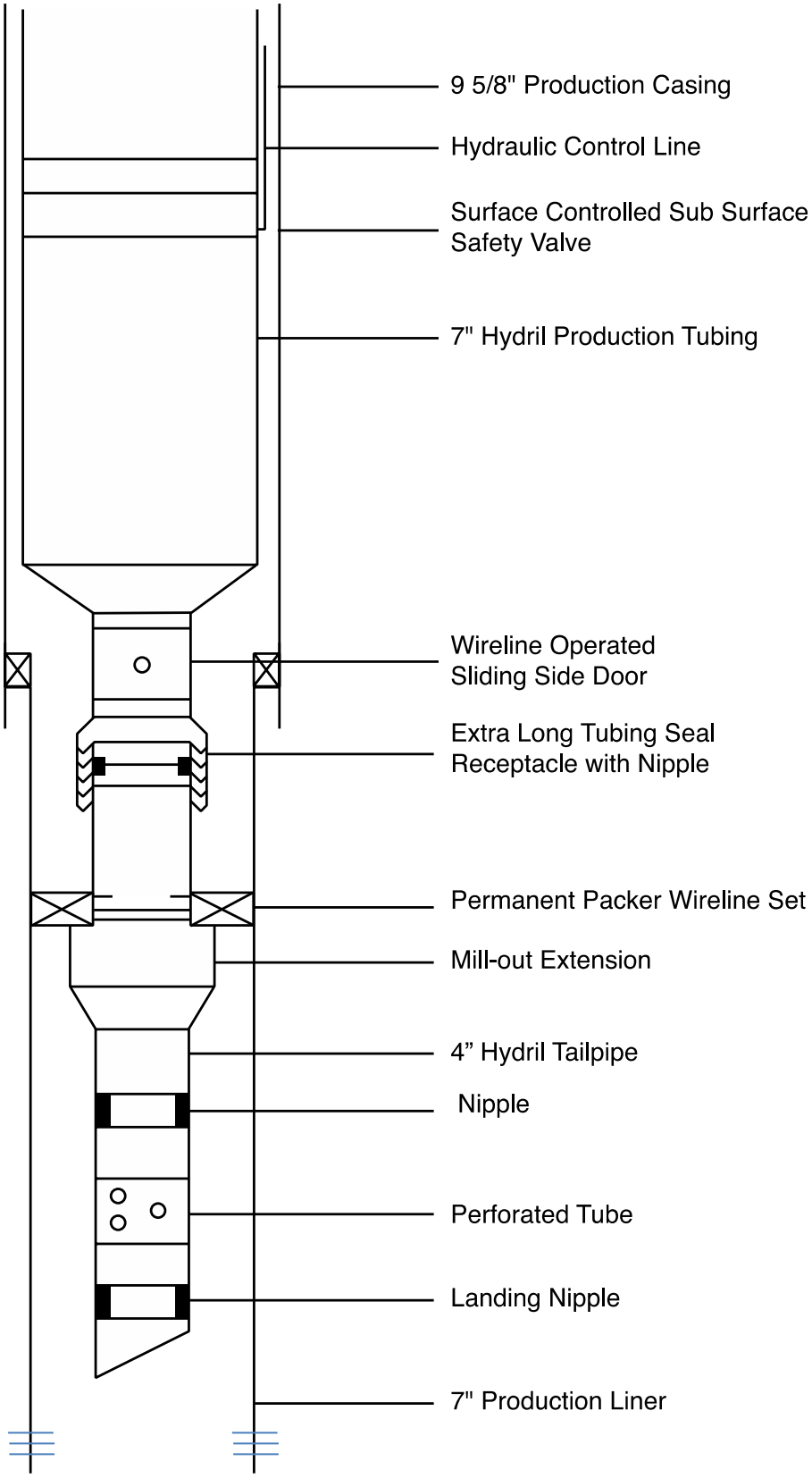

Figure 2: Well completion example 2

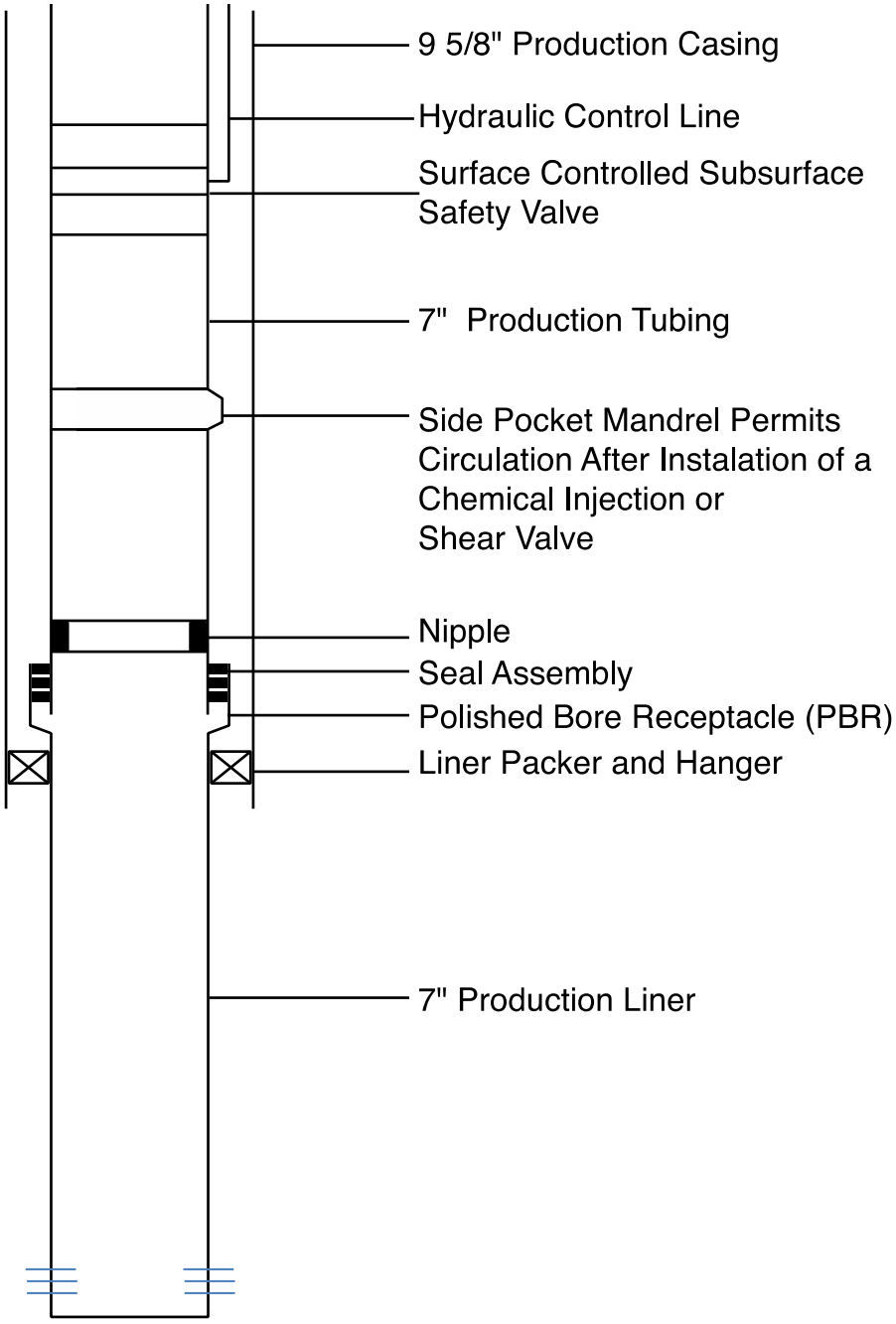

Figure 3: Well completion example 3

NAYEBALE VALENTINE

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G11PT: PRODUCTION TECHNOLOGY -2024-2025

COURSE-WORK

PART ONE.

Question J

i) abrasion and damage of the internal profile of the side pockets by sand particles. Accumulation of sand in the pockets displace of the gas lift valves from their position resulting into reduced artificial lift efficiency.

ii) Sand particles in the wellbore restrict the proper flow of fluids causing a high pressure to drop and an increase in the velocity of the fluids. This changes the flow rate from laminar to turbulent flow, accelerating the abrasion of the internal structure of the side pocket mandrel. The high pressure drops results into the OPR curve shifting to the right and reducing the flow rate.

iii) gas lift valves. They could become stuck due to the build-up of sand particles in the side pocket making them fail to close or open during artificial lift

Question K

i)

- Failure to operate valves when needed to circulate fluids through the side pocket mandrel.
- Sudden changes in the well pressure and reduced efficiency of the artificial lift performance to reduce bottom hole pressure.
- this problem would be detected by regularly reading the pressure gauge to notice any anomalies.
- Regular production logging would show the condition of the completion tools in real-time.

ii)

- flow meter
- noise log

iii) Flow rate: This predicts the flow rate of fluids around the wellbore per depth and could predict a reduced flow rate around the depth where the side pocket mandrel is located.

Noise logs: this tool is used to predict the noises that arise from a leak of fluids in the side pocket mandrel when its damaged.

Question L

i) Scale formation around the wellbore. Calcite minerals precipitate out from of the reservoir fluids when formation waters mix with other fluids injected into the reservoir especially during water flooding. This is due to the reduction of pressure in the reservoir.

ii) Calcite deposits from precipitation clog the pore throats restricting the flow of fluids and forming positive skin. This results into a high pressure drop shifting the OPR curve to the right and decreasing the flow rate.

Question M

i) Acid fracturing. With acid fracturing, the fractures form conductive channels for enhanced fluid flow and acid dissolves the scale, increasing the flow path for fluids and increasing the flowrate.

ii)

- pre-flush stage: A pad fluid is pumped at a pressure higher than the formation pressure to form a fracture within the formation. The fracture made should have a deep penetration that bypasses the formation damage.
- HCL (15- 28%), mixed with polymers is pumped in stages to form un-uniform fractures creating conductive channels.
- The well is shut in.
- post flush: involves pumping more acid for it to flow deeper into the formation.

Question N

Perforation gun: Tubing conveyed perforating guns create deep long channels into the reservoir that bypass the formation damage and increase flow path for fluids.

Perforation charges: Deep penetrating charges of about (12-20) inches into formation

Damage delivery: tubing conveyed perforation. (TCP) This perforation can be done in underbalanced conditions.

Perforation condition: The re-perforation should be done in under-balanced conditions. This helps in the immediate clean-up of the well to remove the scale debris to prevent it from re-depositing around the wellbore.

PART TWO

During oil and gas production, controlling the bottom flowing pressure is important to the type of flow rate desired. A choke is a completion tool used to provide the back-flow pressure to maintain the draw-down in the reservoir required for a high-flow rate. Analysis of the multiphase choke flow model is vital because it predicts the type of flow rate of the reservoir fluids as they flow through the choke and this information is crucial in the optimisation of oil and gas production. The four models discussed below is the hydro model, Sachdeva et al, Perkins and the theoretical slip model.

Sachdeva choke flow model. Paper (1)

A study was done on this model to analyse its accuracy in the performance to predict gas flow rates in oil wells and gas condensate. 512 data sets were used of which 239 were oil well data sets and 273 were for gas. The results in (1) showed that the model was able to predict the gas flow rate accurately in the oil well data sets compared to the gas condensate wells.

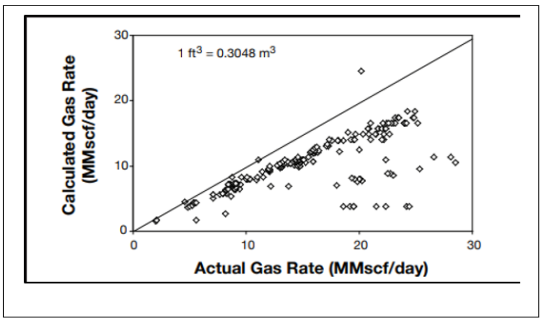

Figure 1 showing predictions for the oil wells

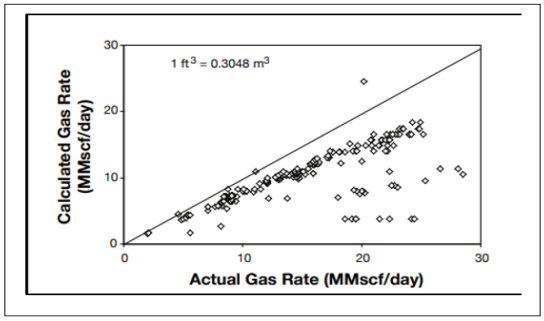

Figure 2 showing predictions for the gas condensate wells

To improve the accuracy of the model, various CD values we used. When the cd values were adjusted to 0.8 and 0.9, the performance showed an error of 15 and 8.5 % which was taken to be in the acceptable range. For the same values used on gas condensate, the accuracy decreases to16% and 21 % respectively, thus, underestimating the measured flow rate.

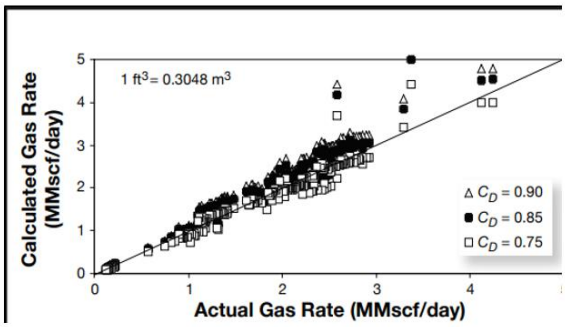

Figure 3 showing results using various CD values for oil wells

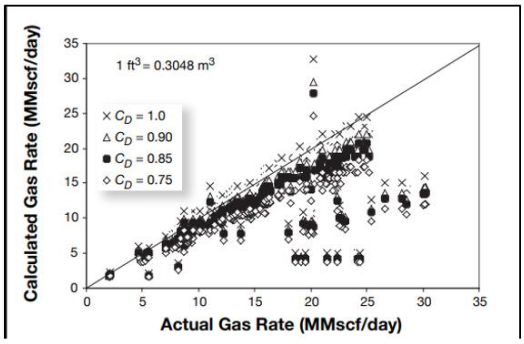

Figure 4 showing results using various CD values for condensate wells

The challenges of this model were that some datasets could not fit int the empirical equation. Mathematically, there were inaccuracies found in finding square roots of negative values leaving out 48 data sets. However, its accuracy increased with the adjustments of the discharge coefficient factor.

<u>Hydro choke model</u>. Paper (2)

The hydro model showed a better performance compared to the Sachdeva et al model with an average error of 6%. They also had a higher performance in the predictions of the sub-critical flow boundary. The Sachdeva model and Perkins model had a higher average error of 2.3% and 32% because they overestimated the single-phase flow-rate data and underestimate a large population of two-phase data.

The high performance of the hydro-model comes from the technical approach of control-volume approach of fluids that analyses the interactions of the multiphase fluids they flow through the choke. It accounts for the pressure losses encountered when two fluids flow through the choke at the same time. It also accounts for the slippage between the phases using slip-correlation, because oil, gas, and water flow at different velocities and have different frictional pressure losses as they flow. The model accounts for all these flow parameters thus have a better performance.

an experiment was carried out in (2). When using field data, hydro model showed a better ability to predict the total mass flow rate compared to Sachdeva and Perkins model.

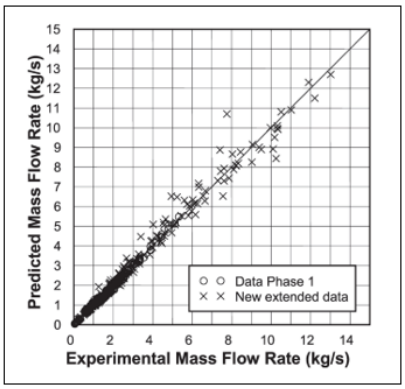

Figure 5 showing predictions of total flow rate by hydro model

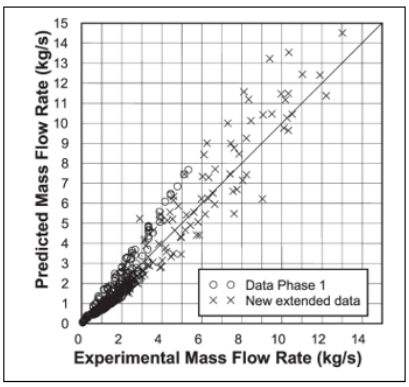

Figure 6 showing predictions of total flow rate by Sachdeva model

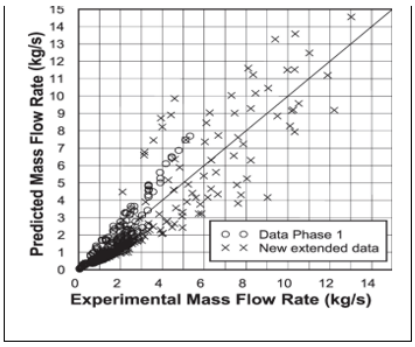

Figure 7 showing predictions of total flow rate by Perkins model

Slip (hydro) model: paper (3)

This model can predict the flow- rate of through choke flow rate prediction complex as they don't flow at the same rate. The model can calculate the total flow rate for both critical and subcritical flow conditions and determine the flow boundary. The design of this model depends on the fluid velocity ratio, if it's high then a high slippage factor is designed for the model.

The slip model out-performed Sachdeva et al model in determining the critical flow boundary and had a lower average error of 5.2% compared to the Sachdeva model whose average error was 19.4%

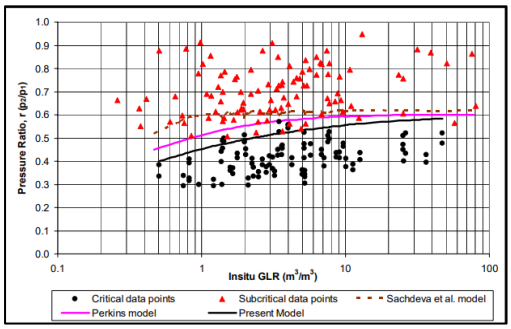

Figure 8 showing the comparison of the models in determining the critical flow boundary

Laboratory test: A validation evaluation is carried in out to analyse the performance of the slip model, Sachdeva at al model and Perkins model in predicting the subcritical flow rate.

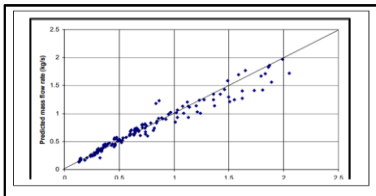

Figure 9: Slip model predictions vs lab measured mass flow rate

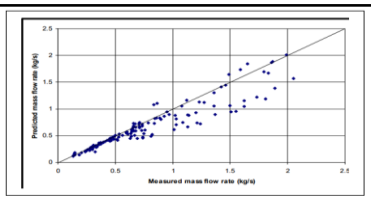

Figure 10: Sachdeva model predictions vs lab measured mass flow rate

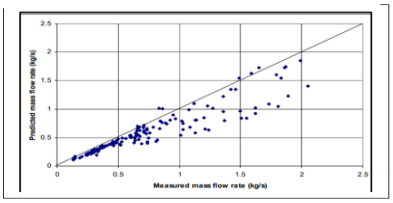

Figure 11: Perkins predictions vs lab measured mass flow rate

The results in (3) show that the slip model had a higher performance with an average error of 2.7% compared to Sachdeva et al model with a 10.7% average error.

Experiment with field data: Middle East and Ashford and Peirce field units were used to compare the results portrayed by the models.

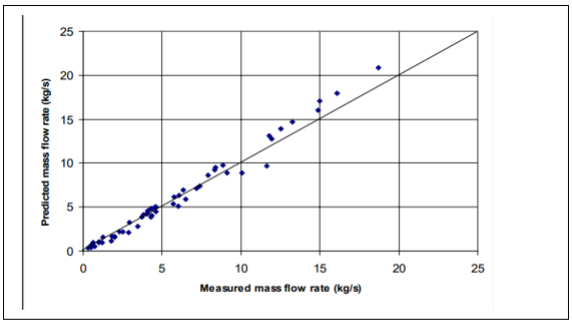

Figure 12 showing slip model predictions vs field measured mass flow rate

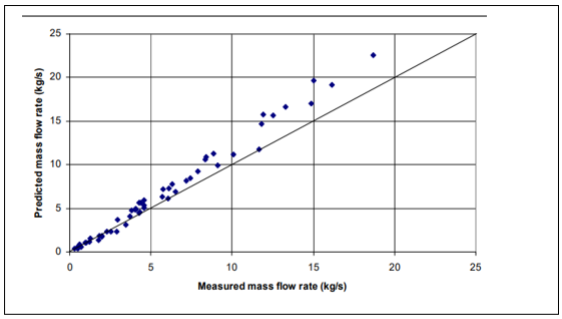

Figure 13 showing Sachdeva et al predictions vs field measured mass flow rate

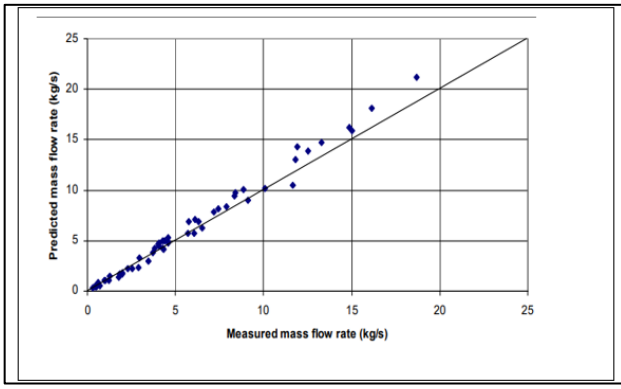

Figure showing Perkins predictions vs field measured mass flow rate

slip model results in (3) showed a high performance with an average error of 1.4%, Sachdeva et al model with 12.5% and Perkins model with 4% average error.

<u>Proposed optimal conditions</u>.

The hydro model shows the best performance compared to the other models because it can work with both single and multiphase flow systems for both critical and sub-critical flow conditions. The combined method approach of control-volume and accounting for the slippage of fluids in multiphase gives the model more strength to predict the mass low rate with low average errors compared to the other the rest of the models.

Discharge coefficient (CD)	0.78 to 1
Temperature (upstream and downstream)	50◦C
Upstream pressure	8 bara
Flow regime	Sub-critical flow conditions

Conclusions

The Sachdeva et al model performs better with adjustments of the CD values. It also predicts the gas flowrate in oil wells better than in the gas condensate wells. The slip (hydro) model performs best in predicting the subcritical and critical boundaries and has a higher performance than Sachdeva due to the approach of accounting for the slippage between phases.

The Perkins model showed the worst performances among all the models with high average errors and underestimation of the mass flow rates.

References

B.GUO, A. A.-B. ,. G., 01/ MAY / 2007. Improvement in Sachdeva's Multiphase choke flow model using field data. *JOURNAL OF CANADIAN PETROLEUM TECHNOLOGY, Issue SPE- PETSOC-07-05-01* **paper (1)**

E.M. AL-SAFRAN, M., 30th/april/2009. Predictions of two-phase critical flow boundary and mass flow rates across chokes. *SPE production and operations,* Issue SPE- 109243-PA. **paper (3)**

R. B. Schuller, s. M. S. S.-o. T., 19/ august/ 2006. Critical and Subcritical Oil/Gas/Water mass flow predictions and experiments on choke. *SPE Production and operations,* Issue SPE-88813-PA. paper **(2)**